

Lab # 1

Plant tissue culture

it's an *in vitro* vegetative propagation technique

you could [↓] ~~you~~

↳ like leave / stem / root

use Flasks or

nodes / internodes found on the stem

tubes or jars

meristematic tissue (نسيج مریستیماتی)

under aseptic conditions

need to use laminar flow = UV light

Each plant tissue culture has to have the following

① preparation room

② induction incubation (growth) room

3 things under control :-

- temperature ($18-22^{\circ}\text{C}$) depends on the plant

dicot plant → need more than it

monocot plant → ($18-22^{\circ}\text{C}$)

- light each plant needs 16 hour

- Humidity 30 - 50%

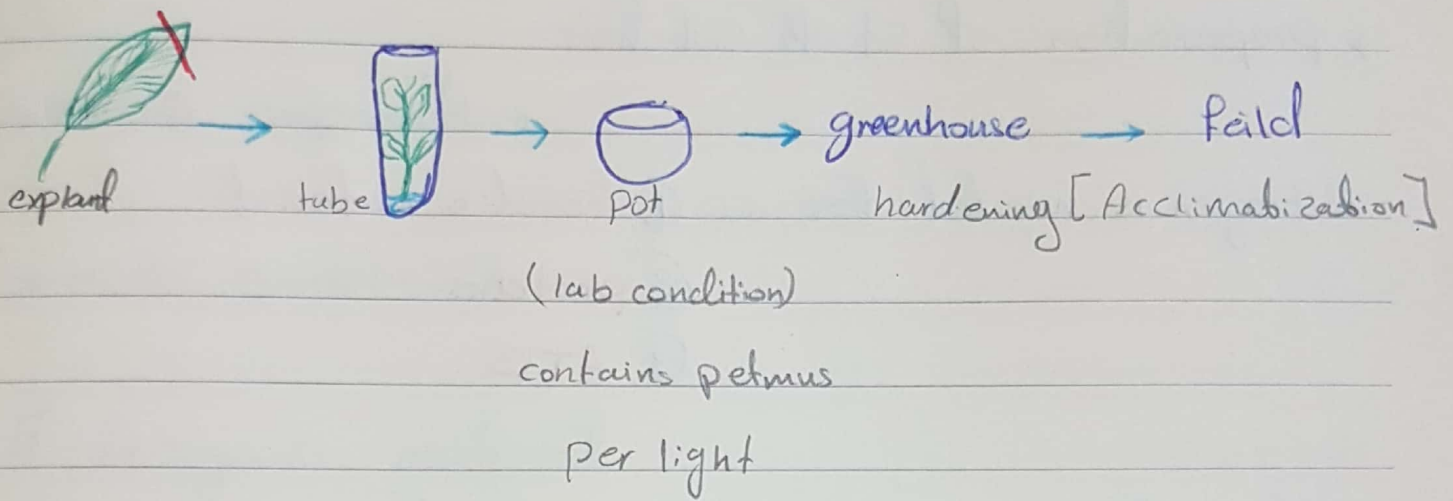
③ transfer room

④ transplantation room

we use laminar flow

sterilized by UV light

④ Transplantation room



Aims or goals of plant tissue culture :-

- ① High Volume Propagation کثیر اور زیادہ
- ② Virus elimination Free of virus
- ③ genotype modification
- ④ stress selection [toxic, drought, salinity]
- ⑤ Germ plasm storage store them in liquid N₂
- ⑥ year round viable
- ⑦ movement of germ plasm across barriers

* yellow and internodes are indication of bad growth

* internodes have no growth therefore we use the node part

* part of the plant has spot of yellow → this means deficiency of sulphur called chlorolysis

Lab # 2

→ preparation of stock solution

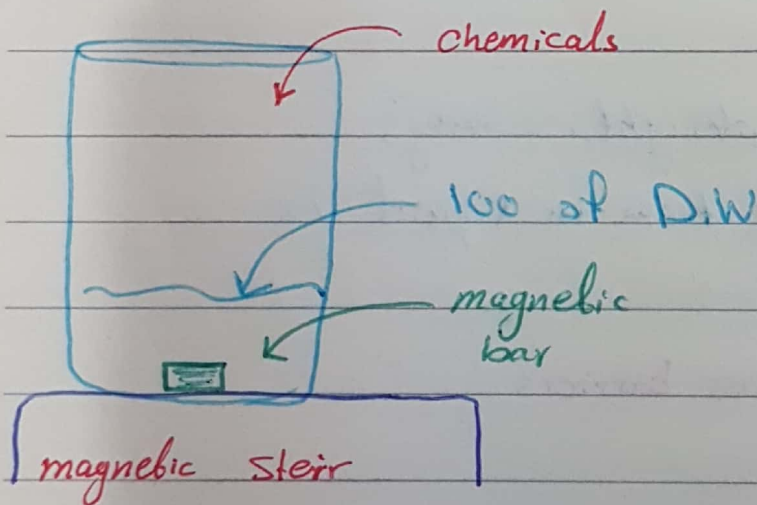
Nitrogen is needed for :-

- ① Genetic material
- ② proteins
- ③ ATP

what bacteria is used in nitrogen fixation?

→

How to prepare a stock solution



* Then add 400 D.W

* put them in brown bottles because the chemicals are sensitive to light to prevent oxidation of chemicals with light

* put in refrigerator to minimize contamination

③

Lab # 3

→ media preparation

* media Constituent

I Inorganic material

macronutrients

Stock I

N → anthocyanic pigment (-ve)

Mg → appear yellow pheophyta (-ve)

P → reduce cambium building (secondary growth)

K → reduce cambium building (secondary growth)

Ca → cell wall synthesis

micronutrients

Stock II

Mo

Mn

Fe

Br

Co

[2] Organic material

Carbohydrates

Source of carbon

(Sucrose)

energy

$$2.5\% \approx 3\% = 30 \text{ g/L}$$

Other than carbohydrates

- vitamins
- amino acids
- coconut milk
- yeast extract

[3] growth hormones for differentiation

- Auxin $\rightarrow$ IAA, IBA

initiate rooting

indole acetic acid $\downarrow$ more stable

To dissolve auxin use normal NaOH

- cytokinin $\rightarrow$ BA initiate shooting

To dissolve cytokinin use HCl

- 2,4-D $\rightarrow$ callus

- G.A₃ $\rightarrow$ cell elongation (80% ethanol)

[4] Vitamins stock IV

initiate growth

[5] PH adjustment

using PH meter

PH for media = 5.7 - 5.9

$$PH = -\log[H^+]$$

a. help in absorption

b. act as a solidifying agent

[6] Agar 0.5% - 0.8% = 8g/L

- gelling agent (solidifying agent)

- Aeration

* media preparation

[1] MS free hormone media

control → salt

[2] Shooting media

MS solution + 0.5 mg/L GA₃ + 2 mg/L BA + 0.3 mg/L IBA

[3] Rooting media

MS media + 0.5 mg/L GA₃ + 2 mg/L IBA

[4] Callus formation (A) (1) 2-4D

(2) combination of auxin + cytokinin

MS media + 0.5 mg/L GA₃ + 0.5 mg/L 2-4D

> 0.5
< 0.5 Toxic

(6)

[5 callus (B)

Ms media + 0.5 mg/L GA₃ + 0.2 mg/L BA + 0.5 mg/L IBA

* preparation of Ms Solution :-

stock I	50 ml	$50 \times 5 \rightarrow 250 \text{ ml}$
stock II	5 ml	$5 \times 5 \rightarrow 25 \text{ ml}$
stock III	5 ml	$5 \times 5 \rightarrow 25 \text{ ml}$
stock IV	5 ml	$5 \times 5 \rightarrow 25 \text{ ml}$

overall volume = 325 ml

1 L

325 ml + 675 D.H₂O

stocks $\frac{1000}{5} \text{ ml Ms} = 200 \text{ ml}$

$\frac{200}{2} = 100 \text{ ml}$

1% $\rightarrow 1 \text{ g} / 100 \text{ ml}$

1g $\rightarrow 1000 \text{ mg}$

1 ml = 1000 micro pipette

1000 mg = 100 ml

2 mg = 0.2 ml

* How to prepare 1 L of Ms media

¹⁰⁰ Ms 200 ml + ⁴⁰⁰ D.W 800 ml + ¹⁵ Sucrose 30 g/L

dissolve ← agar 8 g/L + pH adjustment 5.7-5.9 ←
microwave

→ homogenization by magnetic stirrer → distribution
30 test tube
15 ml / tube

autoclave ← water proof ← cotton plug ←
126 °C / 20 min Paper
1.5 kg/cm²

Lab # 4

Seed surface sterilization

Chick pea seeds

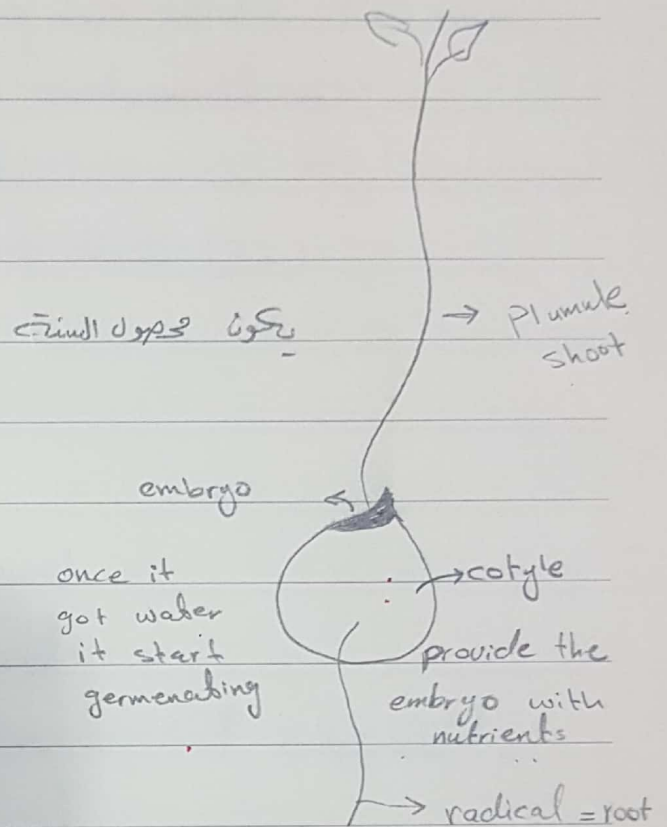
1-2 years

1 Vital seeds

2 germinated seeds

method of sterilization

- 1 methanol bromide → soil
- 2 fumigation
- 3 Heat
- 4 chemicals [Halogens] → all halogens are sterilizers
- 5 Radiations → laminar flow (UV)
- 6 pressure
- 7 Filtration → For heat labile growth hormones



How to sterilize chick pea seed

[1] continuous wash with running tap water / 10 min

→ To clean dust and minimize contamination

[2] wash with soap (detergent) / 10 min

→ breakdown surface tension of H_2O (73)

remove extra soap with H_2O

→ under laminar flow

[3] add disinfectant / 10 min

- Clorox → Sodium hypochlorite

- $AgNO_3$ → very toxic for plant

- 3% H_2O_2 → toxic, carcinogenic chemical compound

[4] Wash with water

[4] Add 70% Ethanol / 30 sec V/V. 30ml/70ml H_2O / ethanol

To avoid dehydration → if increase the embryo will dry (die)

[5] Wash with autoclaved distilled water 3-4 X time / 20 min

[6] Dry under laminar flow

Lab # 5

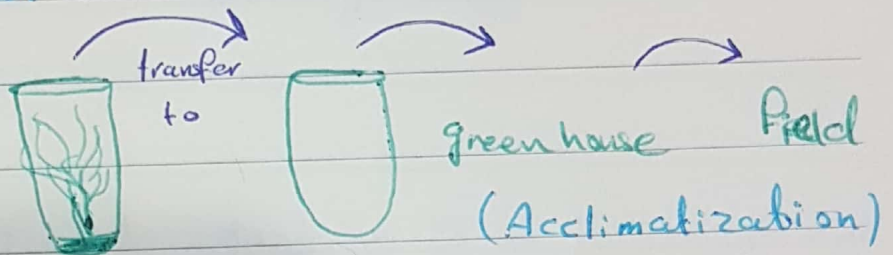
Micropropagation require different techniques

[1] Explant selection leaf, node, internode, shoots

[2] select media (rooting, shooting, callus)

[3] sterilization method

[4] transplantation



[A] multiple shoot induction 3-methods

1- through callus

- indirect method

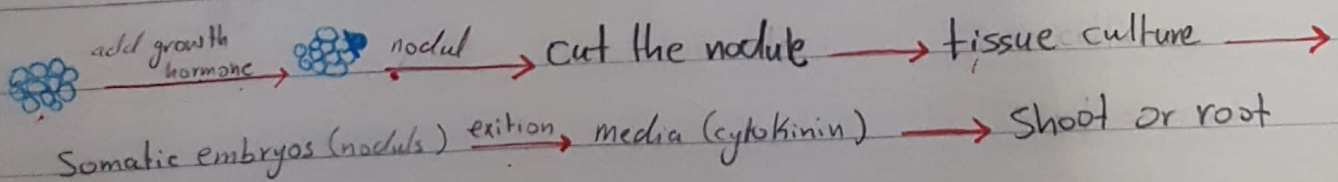
@ pass through callus undifferentiated mass of cells

explant $\xrightarrow[\text{auxin \& cytokinin}]{\text{growth factor 2-4D}}$ explant undifferentiated genetic variation

meristematic tissue $\rightarrow$ genetic variation

(b) Organogenesis most efficient to produce callus

Some shoots unstable



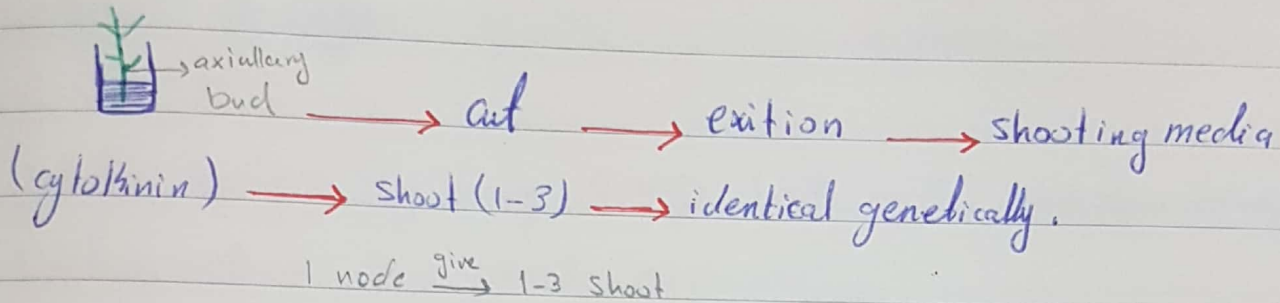
use vegetative propagation → to get identical (stable) genetic material

we prepare 5 types of media → To study the response of plant to different types of hormones (phytohormones)

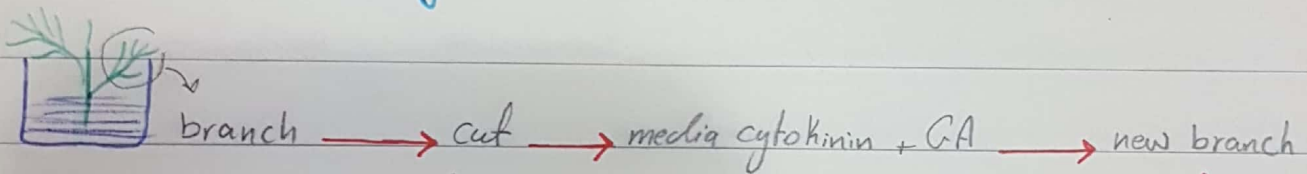
nodule → unfertilized ova

2. Enhancing of axillary bud (direct method) الثقة الكرونية

- no pass through callus
- no genetic variation



3. Enhancing of adventitious branching: لازم يكون عدد 3 عقد كرونية combination of growth hormones



[B] Root induction.

media (Auxin) IAA, IBA, NAA ^{more stable}

* Vegetative propagation [to keep genetic material constant]

- Callogenesis Formation of callus by → 2.4D

تكوين الكالوس

→ cytokinin + auxin

* Callus induction :-

Types of callus :- [1] friable callus [loosely attached]

- very important in cell culture
- give somatic embryos [unfertilized ova]

[2] compact callus

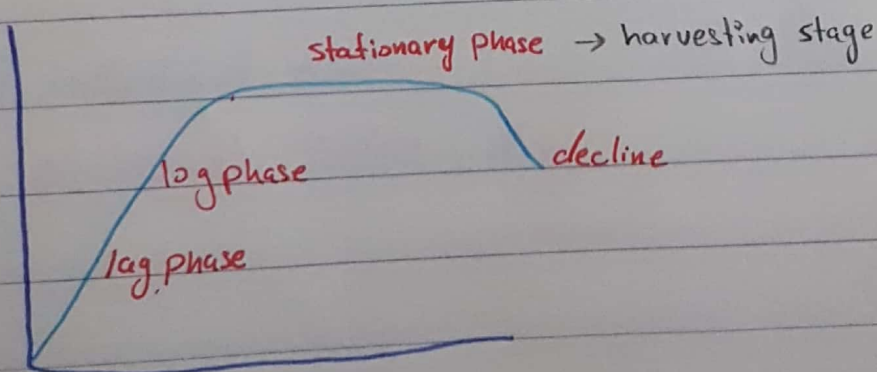
Friable callus $\xrightarrow{\text{Prepare}}$ cell culture $\xrightarrow[\text{need}]{\text{For growth}}$ liquid form of media

for callus induction use :- single cell [not 100% success]

• multi cells ($10^4 - 10^5$) $\rightarrow$ مجموعه الخلايا

$\rightarrow$ Some cells secrete stimulators to stimulate other cells to grow

cell culture growth pattern $\rightarrow$ Similar to prokaryotic growth pattern



• S-phase

• Sigmoid curve

How can I know if my culture is not contaminated??

→ turbidity → contaminated [there is bacteria]

1st part
 [4] DNA manipulation DNA changing by physical or chemical
 DNA → phenotype on the phenotype level

a. as transformation

- pure DNA $\xrightarrow{\text{transfer}}$ plant cell - direct
- indirect
- use ions or chemicals
- * Ca^{+2}
- * DMSO dimethyl sulfoxide
- * PEG polyethylene glycol
- * liquid N_2

b. as transduction

- direct
- through virus like lysogenic

c. electroporation

most efficient

- direct
- through electrical current

d. direct gene gun



particles with DNA
shooting

e. direct injection

microscopic needle

How to isolate protoplast?

isolation mechanism

[1] chemical 10⁶ cells high yield cell wall protoplast washing
enzymes - cellulose
 - pectinase

[2] mechanical buffer 1 cpw

Advantages

no effect of harmful enzyme

no effect of osmotic pressure

disadvantage

low yield

mechanical methods:-

• plant tissue (20g)

- remove midvein (midrib) → Vascular tissue [Xylem & dead tissue] not required

* Surface Sterilization (under laminar flow)

— clean from mud and dust

* Peel lower epidermis . why?

spongy layer cells are loosely attached and have intercellular spaces that help the entrance of enz & cpw

* Fragmentation

cut tissue into smaller fragments to increase the surface area

* Grinding

using pestle and mortar

هرس مو دق (سحق)

add cpw + 10% manitol to make ⁽¹⁾hypertonic conditions and increase ⁽²⁾osmoticum

* incubation for 20 min at room temperature
to allow the cpw and manitol to penetrate cells

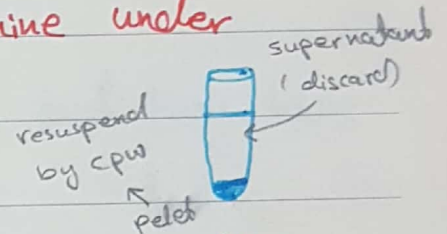
* purification (filtration) using cheese cloth (gause)

* centrifugation 3000 rpm / 10 min

* prepare wet mount slide and examine under microscope

- spherical shaped light green

- if cells were dark make dilution with buffer



chemical method :- في اناء ميكانيكي

- remove cpw → Flood the tissue with enzyme mixture

- Pectinase or any enzyme is dissolved in buffer

- incubate for 2 hrs

- purification

lab # 76

Transformation of carrot slice with *agrobacterium tumefaciens*

Transformation is transferring of DNA from an organism to another

- Carrots are roots, we need to cut them
- we can use eggplant, potato, etc. but they have more water content.

Transformation

1. direct

2. vectors

plasmid

Ti plasmid

tumor inducing plasmid

Ri plasmid

root inducing plasmid

viral → From virus

Ti-plasmid [How Ti-plasmid mediated plant transformation?

Natural Genetic engineered

bacteria → *Agrobacterium tumefaciens* → cause crown gall disease

① Soil microbe

② Gram -ve

③ has extra chromosomal DNA known as Ti-plasmid

- Structure of Ti-plasmid

- ① double strand DNA
- ② close circular DNA

- parts of Ti

الاجزاء

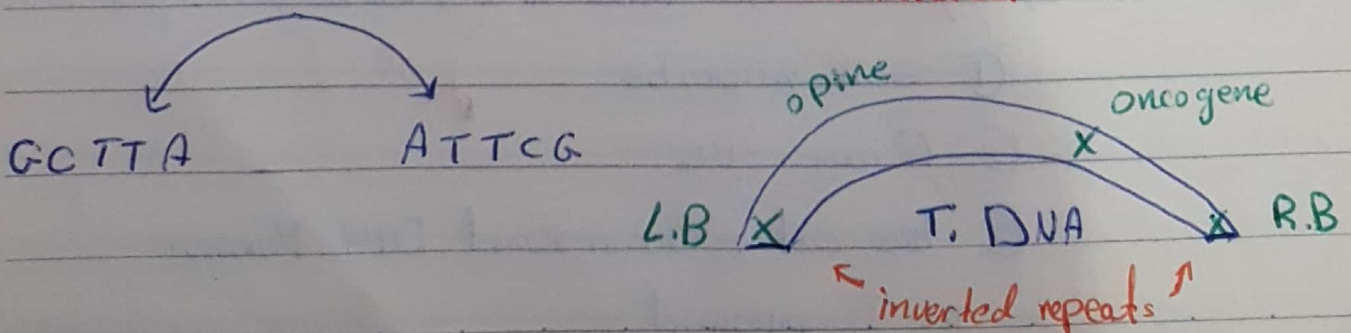
- ① T-DNA → Causes plant cancer ← سبب السرطان
- ② Virulence gene
- ③ origin of replication

☐ T-DNA transfer from bacteria to host [plant cell] and expressed into tumor formation.

- oncogene → pathogenic part, cause plant cancer

translate growth hormones (auxin & cytokinin) → ↑ cell, ↑ mass

named armed gene that protect → ~~new~~ disarming - removal of oncogene & insert gene of interest.



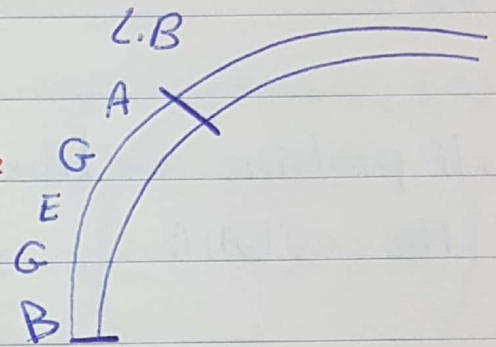
- opine s. marker gene, group of unusual amino acid used to selection process

→ deletion of R.b, and outer layer removal

→ L.B makes the insertion of the gene of interest, there are similarities between T-DNA and plant genome that makes the insertion easy

* Vir genes: located outside T.DNA

Gene locus → at least 10 Vir genes are going to be translated into proteins



VirA → do not need other genes to stimulate / sensory gene / act as signal / continuously produced

• we need wound (scratched) plant

• root is used → lives in soil, agrobacterium lives in soil

→ genomes are similar

- wounded plant $\rightarrow$ secrete phenolic compound (benzene ring + OH)

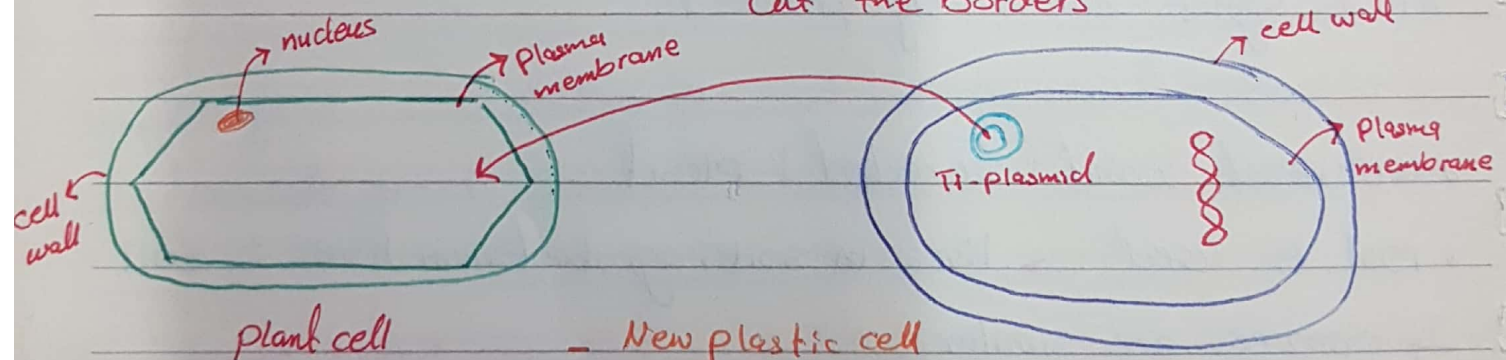


- chemical attraction for bacterial cells
- VirA + Phenolic compound $\rightarrow$ stimulate (active) VirG $\rightarrow$ 2 of the Vir-genes act as endonuclease and stimulate VirE

- 11 proteins are translated from VirE & act to coat the ssDNA

- * VirG $\rightarrow$ produce proteins that act as endonuclease

cut the borders



plant cell
(normal)

- New plastic cell

- cancer cell

bacterial cell

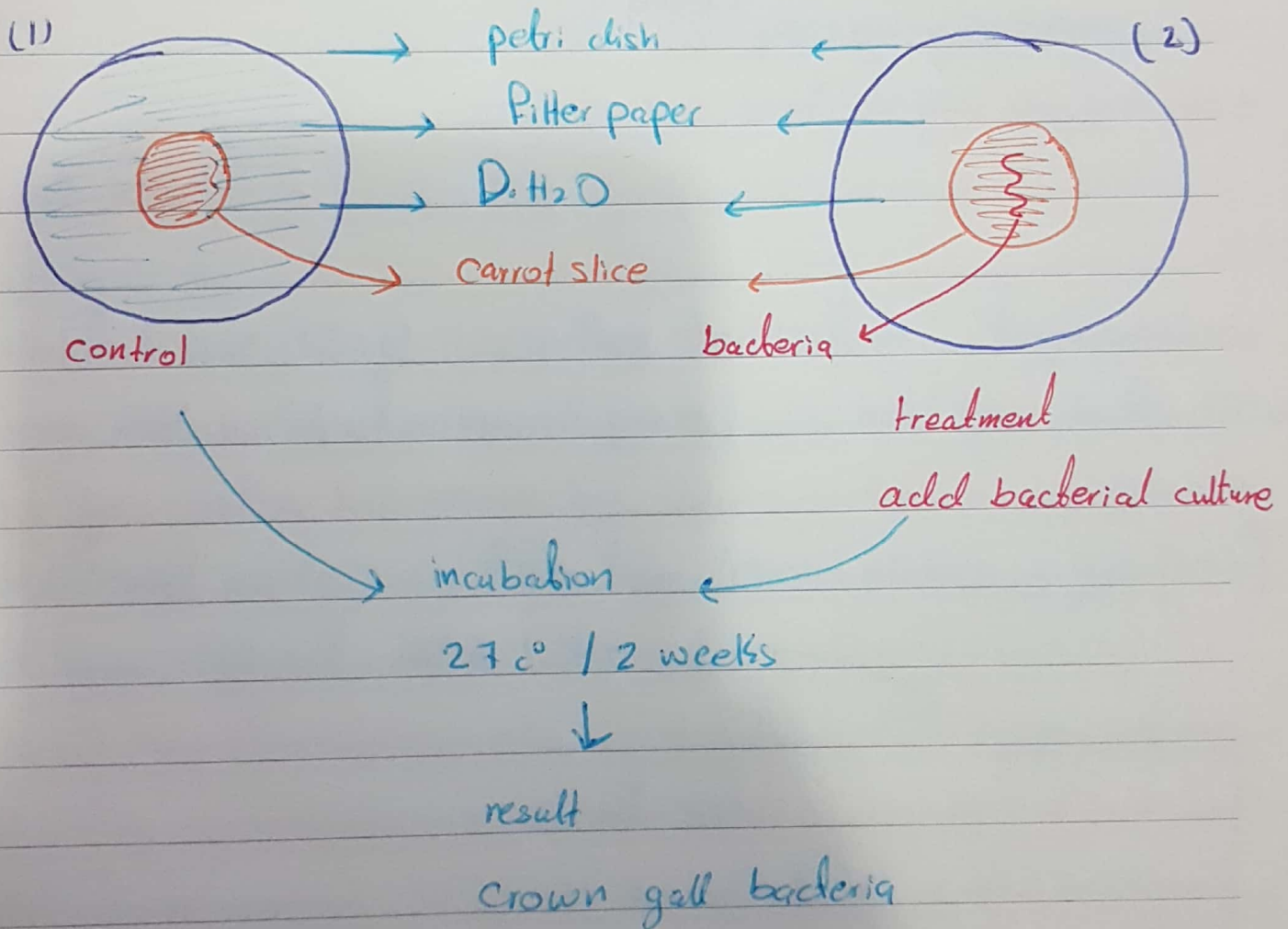
- tumor cell

transformation

* Lab Work :-

Plant tissue $\rightarrow$ Surface sterilization $\rightarrow$ cut into slices
 carrot
 (in vivo)

For each group $\approx$ petri dish



Lab #8

extraction of DNA

↳ in the nucleus

* DNA also found in all organisms but in small amount.

cell lysis → cell opening → cell wall
damage → plasma membrane

* على الأقل بدري امصل على 50ml في ال extraction

Lab work

سپانچ

plant tissue (20g) → surface sterilization → fragmentation
remove mid vein (cut into small pieces) → add lysis buffer (20 ml)

This lysis buffer has phenol SO لا تخلط مع ايدىك → grinding (pestel and mortar) → filtration (cheese cloth) → get (50 ml) of lysis extract add it to

crude extract contain

every thing fat, protein....

all of them toxic

carcinogenic [most toxic - 24 ml chloroform]

phenol, chloroform ...

less toxic is SDS]

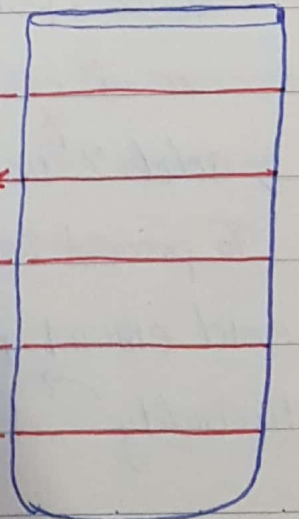
• 10 ml SDS

- 1 ml iso amyl alcohol

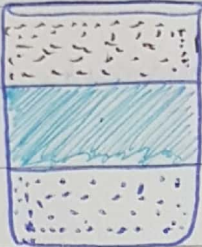
• 24 ml chloroform

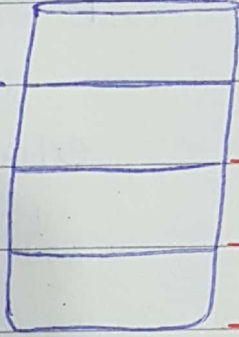
- 25 ml phenol

• 50 ml lysis extraction

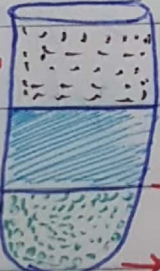


→ shake by hands for 10 min → centrifugation 4000 rpm / 15 min
 ↳ at room temperature

→  → upper (aqueous layer, DNA) brownish color → pipette out
 → middle layer (protein) milky in color
 → lower layer (phenol, chloroform) debris

→ aqueous layer (DNA) add to 
 1 ml isoamyl alcohol
 24 ml chloroform
 25 ml phenol
 باقي SDS لأنو تخلصها فإغلب كمية
 Protein في phenol, proteins في

→ shake → centrifuge again (4000 rpm / 15 min)

→  → DNA
 → protein → pipette out the aqueous layer (DNA)
 → pellet put it in graduated cylinder
 brownish
 white
 dark green

40 ml cold ethanol

→ add 2[↑] volume of cold ethanol (95%) + 3:60 1:20 K-acetate

To prevent DNA degradation by enzyme [inhibit the enzyme activity]

cold ethanol + K-acetate → precipitate DNA → bubbles + turbidity (DNA is formed)

→ swirl and spool chromosomal DNA → Fibrous DNA

* we have 3 reagent to precipitate the DNA

[1] Ethanol we will use it

[2] isopropanol

[3] K-acetate we will use it

* Depends on the DNA size :-

DNA: -ve charge

Small DNA in size → ethanol precipitation according to charge

large DNA in size → isopropanol precipitate large DNA

* extract → we can call it crude extract → it contain every thing (fat, protein)

SDS + phenol → hydrolysis for proteins

To remove the proteins we add proteinase

in this experiment we will use phenol + SDS to remove the proteins not the proteinase But this won't remove all the proteins so we will do centrifugation

→ now we have phenol in our sample so we add chloroform to get rid off the phenol

→ we remove the phenol by chloroform But we have the chloroform in our sample so we remove it by adding 1 ml of isoamyl alcohol and this isoamyl alcohol can evaporate

Lab 9

Hot Alkaline method for Ti-plasmid isolation

Alkaline :- Denaturation for protein

- Denaturation for DNA

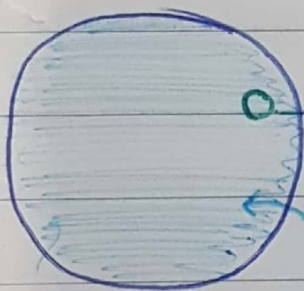
L → pH = 7 - ↑
 L → optimum pH = 8-12.6

cell lysis :- cell opening

- damage cell wall

- damage organelles

Lab work

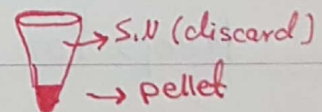


Agrobacterium tumefaciens

Add 2ml of st. D.H₂O

L → pipette bacteria to abendrof tube (2ml)

take pellet and add 1ml of solution I ← centrifuge 4000 rpm / 15 min
 of solution I

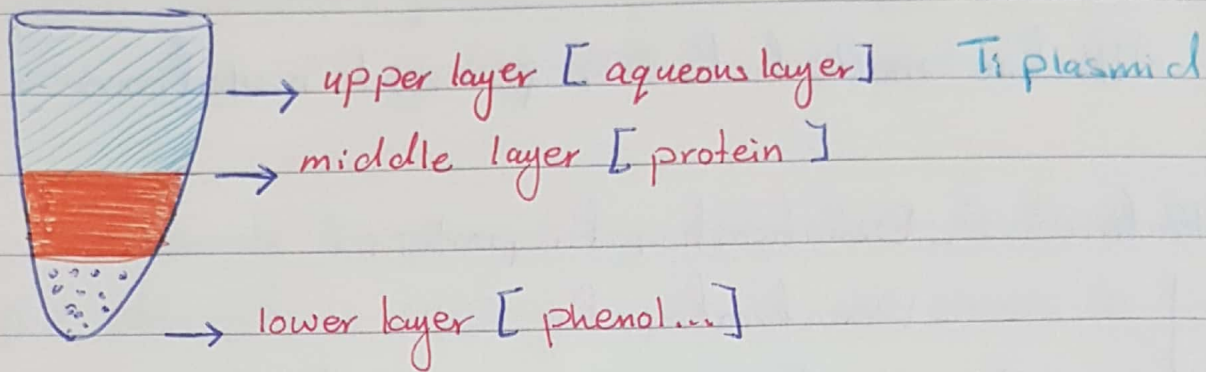


mix

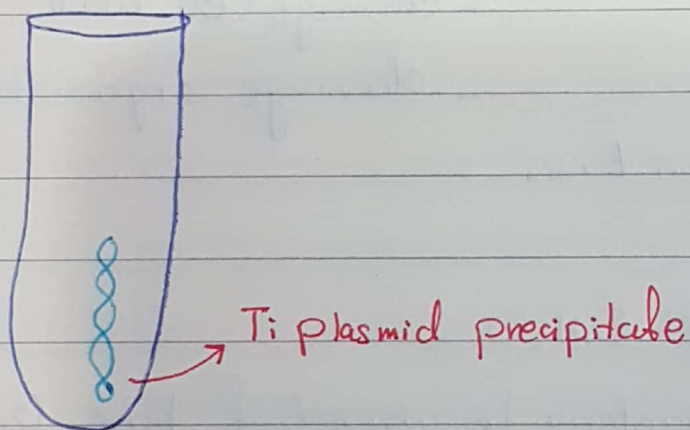
L → vortex → add 2ml of solution II
 - viscous
 - cloudy

add 1st volume of phenol & chloroform (4ml) ← incubation over 60°C / 30 min

vortex mix → fill 4 abendrof tubes → centrifuge 5000 rpm / 10 min



↳ take the aqueous layer and add 2 volume of cold ethanol for precipitation



Keep for next lab for gel electrophoresis